

AKADEMIKA

DCL-08

**PAM / PWM / PPM
MODULATION & DEMODULATION
KIT**

EXPERIMENTAL MANUAL

... .A MESSAGE FROM

Today's system designers are faced with tomorrow's problems. DIGITAL COMMUNICATION is one of the important subject need to teach while learning electronics.

It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO;

-Light years ahead – refers to leadership.

As leaders in our industry in India, we are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have a complete, truly innovative & superior training product. We are so committed to quality that we back our products with a complete comprehensive warranty.

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SAFETY RULES

Carefully follow the instructions contained in this manual as they provide you with important points on safety during the installation, use and maintenance. Keep this manual always with you for easy reference.

Arrange all accessories in order after unpacking, so that their integrity is checked with respect to the packing list. Also ensure that no visible damage as such appears on any accessories.

Before connecting the power supply to the kit, be sure that the jumpers and connecting chords are connected correctly as per experiment.

In DCL-08 kit, signals are to be monitored with an oscilloscope as explained in the manual. For observation of signals on oscilloscope either use X10 (Attenuation Probe) or 180Ω resistance in series with normal oscilloscope probe.

This kit must be employed only for the use for which it has been conceived, i.e. as educational equipment and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of any fault or malfunctioning in the trainer kit, turn off the power supply and do not tamper the kit. In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting chords, Crystals, etc. and other imported items.

This warranty is subject to the following conditions and limitations. The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika Lab Solutions sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the warranty period if the service is needed, purchaser should get in touch with the service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solutions judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and upgradation of products, be sure to contact our service center or sales outlet.

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TECHNICAL SPECIFICATIONS

Signals

Internal frequency: Variable -- 1Hz to 30Hz; 0 - 2 Vp-p (a.c.).

Fixed -- 500Hz and 1 KHz; 0 - 4 Vp-p (d.c. shifted).

External frequency: Up to 4KHz. (0-4 Vp-p).

Sampling

Internal sampling clock: 8 KHz, and 16KHz. with 50 % Duty Cycle.

Modulation Techniques

Pulse Amplitude Modulation.

Pulse Width Modulation.

Pulse Position Modulation.

Voice Communication

Voice link using Noise and Audio Pre-Amplifier and Audio amplifier kits (not supplied with DCL-08 kit alone).

Low Pass Filter

4th Order Butter worth filter with cut -off frequency of 3.4 KHz.

Switch Faults

8 Switch Faults are provided onboard to study different effects on circuit.

Interconnections

2mm sockets & connecting chords are provided for connections on board.

Test Points

All relevant test points are brought out for observation. Observations are carried out on an oscilloscope.

Power Supply

-12V, +12V, +5V, GND

Accessories

DCL -08			Quantity
1	Mini Links PCS2 Patch cord	Male To Male	6
2	Jumper Caps Small (5 mm)		5
3	DCL-08 EXPT Manual		1
4	Power Supply		1
5	Power cord		1

FEATURES

Sine Wave Generation

This section basically provides two Amplitude variable (0-4Vpp) synchronized sine waves each of 500Hz and 1 KHz. The 6.144 MHz crystal oscillator generates a 6.144 MHz clock. It is divided by 2 decade counter and 2 ripple counters to get the 16 KHz, and 8 KHz clock. This signal is fed to serial to parallel shift register, which generates stair case sine wave, by serial shift operation. The serial to parallel shift register has a resistive ladder network at the output. For 16 shifts of the register, one stair case sine wave is produced. So if a 16 KHz clock is fed to the shift register, 1 KHz sine wave is generated. Thus by using the 16 KHz and 8 KHz clocks, sine wave of 1 KHz and 500 Hz are generated respectively. The active filter at the output suppresses the ripple and also takes care of the impedance matching. Filtered output gives smooth sine waves. Low frequency (1-32Hz) signal is generated using IC XR2206.

Sampling Clock Generation

This section provides two different sampling frequencies of 8 KHz and 16 KHz. All these frequencies are generated through main clock of 6.144MHz. The different sampling clock can be selected by jumper JP1 provided on the kit.

Pulse Amplitude Modulation Section

This section provides the pulse modulated signal output at the PAM OUTPUT Post. The pulses are flat top sampled.

Pulse Amplitude Demodulation Section

Flat top sampled PAM signal can be passed through 4th order butter worth filter. It gives demodulated output at filter output post, which is same as original signal.

Filter Section

4th orders Butter worth Low Pass Filter have been provided which filters out the sampling frequency components from the samples and recovers the base band signal.

Pulse Width Modulation Section

The PWM modulator is designed around PWM chip SG3524. The PWM Modulated output is available at the PWM OUT Post.

Pulse Width Demodulation Section

This section consists of a counter and DAC chip. The train of varying pulse width gives varying count values and accordingly DAC gives output, which is directly proportional to the amplitude of input signal. This is then filtered to recover original signal.

Pulse Position Modulation Section

This section uses a single CMOS IC 4098 which is dual monostablemultivibrator. The position of the TTL pulses is changed on time scale according to the variation of input modulating signal, which is PWM output.

Pulse Position Demodulation Section

This section uses an OR gate and a flip-flop. Since PPM corresponds to the end of PWM pulse, output of flip-flop is exactly same as that of PWM signal. This signal is then demodulated using the same technique as that of PWM demodulation.

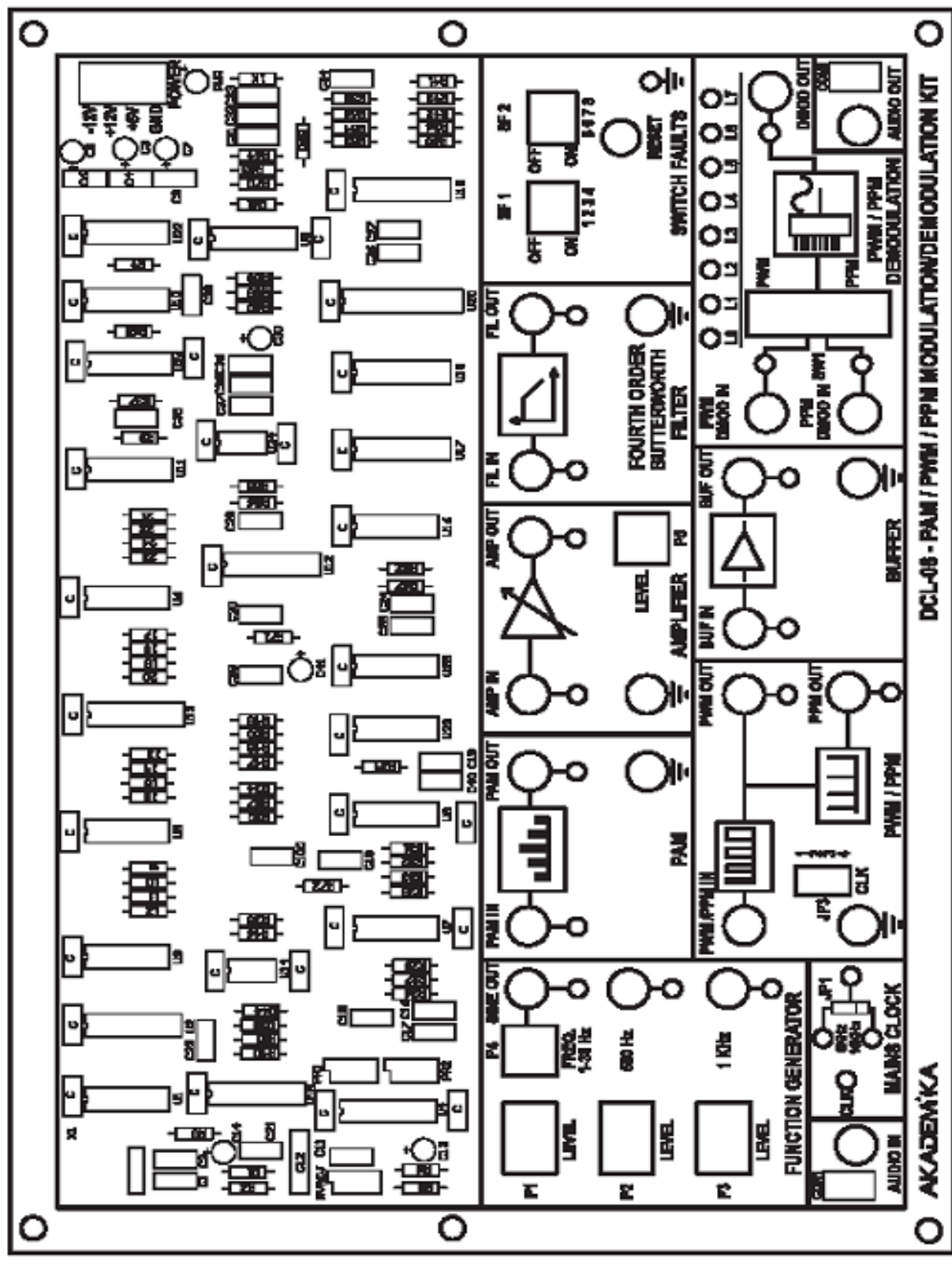
Amplifier Section

This section provides amplification of analog signals and the gain of the amplifier can be varied using potentiometer.

Switch Faults

There are 8 switch faults on the module. SF1 is short circuit module and SF2 is open circuit module. By setting the switch on different position students can observe the different effects on waveforms. Refer to the circuit diagram.

FUNCTIONAL BLOCK



INTRODUCTION

“DCL-08: PAM-PWM-PPM MODULATION TRAINER KIT” is a single board unit in wooden casing and it provides hands on experimentation in the area of Pulse Modulation and Demodulation Techniques such as Pulse Amplitude Modulation (PAM), Pulse Width Modulation (PWM) and Pulse Position Modulation (PPM).

An onboard sine wave generator provides two synchronized amplitude variable sine waves of 500Hz and 1 KHz and one synchronized amplitude (0-2vpp) & Frequency (1-30Hz) variable sine wave. The unit has provision to accept external sine wave as input. An onboard clock generator circuit provides clock frequency of 8 KHz and 16 KHz for sampling circuit.

With this kit one can study Pulse Amplitude Modulation / Demodulation, Pulse Width Modulation / Demodulation and Pulse Position Modulation / Demodulation. One can also perform voice communication using above Pulse Modulation & Demodulation techniques using Noise and Audio Pre-Amplifier and Audio amplifier kits (not supplied with DCL-08 kit alone). All onboard signals can be easily interconnected with the help of connecting chords. The generated signals are phase locked to enable the learner to observe stationary signals on oscilloscope. Test points are brought out on board & identified. The trainer has a complete circuit mimic inscribed on the PCB.

• Principles Of Pulse Modulation

Pulse Modulation may be used to transmit analog information such as continuous speech or data. Here continuous waveforms are sampled at regular intervals. Information regarding the signal is transmitted only at the sampling times, together with the synchronizing pulse that may be required. At the receiving end, the original waveforms may be reconstructed from the information regarding the samples, if these are taken frequently enough. Despite the fact that information about the signal is not supplied continuously as in AM and FM, the resulting output at the receiver can have negligible distortion.

Pulse Amplitude Modulation (PAM)

Pulse Amplitude Modulation is a technique of communication in which, high frequency square wave is modulated by low frequency analog signal. The pulses sample the modulating signal. The PAM signal is nothing but the high frequency square wave in which the amplitude of each pulse is equal to that of the information signal at that respective sampling instant.

Pulse Width Modulation (PWM)

This technique of modulation controls the variation of duty cycle of the square wave (with some fundamental frequency) according to the modulating signal. Here the amplitude variation of the modulating signal is reflected in the ON period variation of the square wave. Hence it is also called as technique of V to T conversion.

Pulse Position Modulation (PPM)

In this technique of modulation, the position of TTL pulse is changed on time scale according to the variation of input modulating signal amplitude. The pulse positions are directly proportional to the instantaneous values of the modulating signal.

• Circuit Description

Sine Wave Generation

This section basically provide two Amplitude variable (0-4Vpp) synchronized sine waves each of 500Hz and 1KHz. The 6.144 MHz crystal Oscillator generates a 6.144 MHz clock. It is divided by 2 decade counters U2 (74LS390) and 2 ripple counters U3 (74LS393) to get the 16KHz and 8 KHz frequencies. This signal is fed to serial to parallel shift register U5, U6 (74HC164), which generates the sine wave, by serial shift operation. The serial to parallel shift register has a resistive ladder network at the output. For 16 shifts of the register, one sine wave is produced. So if a 16KHz clock is fed to the shift register, 1 KHz sine wave is generated. Thus by using the 16KHz and 8 KHz sine wave of 1KHz and 500Hz are generated respectively. The active filter at the output suppresses the ripple and also takes care of the impedance matching.

Variable Frequency sine wave is generated around U4 (XR2206). For Frequency variation P4 Potentiometer & for Amplitude variation P1 Potentiometer is used.

Sampling Clock Generation

This section provides two different sampling frequencies of 8KHz and 16KHz. All these frequencies are generated through main clock of 6.144MHz. The different sampling clock can be selected by jumper JP1 provided on the kit.

Pulse Amplitude Modulation Section

Refer to the circuit diagram as shown later in this manual sheet 2 of 5 of **DCL-08**. This section provides the pulse modulated signal output at the PAM OUT Post. The Pulse Amplitude Modulation circuitry is built around the analog switch U11 (CD4066). The sampled signal is obtained at the output of PAM Modulator. The amplitudes of the regularly spaced rectangular pulses vary in direct proportion to the instantaneous sample values of a continuous analog signal. One can select the sampling frequency clock of 8 KHz and 16 KHz from onboard clock generator.

Pulse Amplitude Demodulation Section

PAM signal can be passed through 4th order low pass butter-worth filter U9 (KF347). It gives demodulated output as per original signal at FILTER OUT post. The Pulse Amplitude Demodulator consists of 4th order Active Low Pass Butterworth filter. This filters out the sampling frequency and their harmonics from the modulated signal and recover the base band by integrated action. The cut-off frequency of the 4th order low pass butter-worth filter is 3.4 KHz.

Pulse Width Modulation Section

The PWM modulator is designed around PWM chip U15 (SGS3524). The PWM Modulated output is available at the post PWM OUT. This technique of modulation controls the variation of duty cycle of the square wave (With some fundamental frequency) according to the input modulating signal. Here the amplitude variation of the modulating signal is reflected in the ON period variation of square wave. Hence, it is a technique of V to T conversion.

The circuit is divided in two stages. First stage is the circuit built around Op-Amp LM741. This is unity gain non inverting amplifier designed as level shifter. When there is no input signal, the output of the amplifier simply follows the voltage at inverting input terminal, with inverted polarity. This is necessary because PWM chip requires only positive polarity signal. Now the input signal fed to non-inverting terminal over rides this shifted voltage.

This signal is now fed to second part of the circuit. This is specialized for regulating Pulse Width Modulation. This primarily consists of one fixed frequency oscillator whose

frequency is decided by timing resistor R_t and capacitor C_t . The regulation is simply given by formula.

$$F = 1.18 / (R_t \times C_t)$$

Where F is in KHz, R is in Ω and C is in μF . Also an error amplifier, comparator and flip-flop are important functioning parts internal to the IC. When no input signal is given to the error amplifier the output oscillates with the same frequency decided by the external R & C network. When input is given to the error amplifier, it causes a linear charging of the output pulse. The amplifier output is compared with the linear charging voltage at C_t resulting in a modulated pulse at the output of high gain comparator. Thus we get the PWM. For more details please refer to the appendix SGS3524 data sheets.

Pulse Width Demodulation

The input signal is Pulse Width Modulated, so the ON time of the signal is changing according to the modulating signal. In this demodulation technique, during the ON time of PWM signal one counter is enabled. At the end of ON time counter U17 and U18 (74LS191) outputs a particular count, which directly corresponds to the amplitude of input signal. Then this count is fed to a DAC U19 (DAC0800). The output of DAC corresponds to the amplitude of input signal. Thus train of varying pulse widths gives varying count values and accordingly DAC gives outputs, which is directly proportional to amplitude of input signal. This is then filtered to get original signal. Thus, at the output we get the original modulating signal extracted from PWM wave.

Pulse Position Modulation Section

This section uses a single CMOS IC 4098 which is dual monostablemultivibrator. The position of the TTL pulses is changed on time scale according to the variation of input modulating signal. The ON period of the output pulse from the monostable is decided by the external R and C network. Here, the active low reset input pin (3) is tied to VDD. Circuit uses monostable 1 in the IC. Pin 4 and 5 are the rising and trailing edge trigger inputs of IC. Here trailing edge triggering is used. Output is taken from pin No.6 of monostable 1 (Q).

Now Pulse Width Modulated signal is fed as input to this circuit. Please note that input modulating signal must be converted into Pulse Width Modulated form before applying to pulse position modulator. As the signal is PWM, naturally, according to the input signal, the pulse duration is changing and this change in pulse duration causes delay in triggering. The input is given to trailing edge trigger input of monoshot. So finally we get the pulses at the output, which are shifted on the time slot. This is nothing but pulse position modulation.

Thus Pulse Positions are directly proportional to the instantaneous values of modulating signal.

Pulse Position Demodulation Section

This Pulse Position Modulated signal is ORed with U10 pulse, generated by the rising edge of pulse width modulated signal. The output of the gate is fed to CLK input of U22 (74LS74) flip-flop. Thus flip-flop acts as a bistablemultivibrator giving out high output for the duration between rising edge of PWM signal & PPM signal. Since PPM corresponds to the end of PWM, pulse output of flip-flop is exactly same as that of PWM signal. This signal is then demodulated using the same technique of PWM demodulation.

Amplifier Section

This section provides amplification of analog signals with help of U9 (KF347) and the gain of the amplifier can be varied using potentiometer P5.

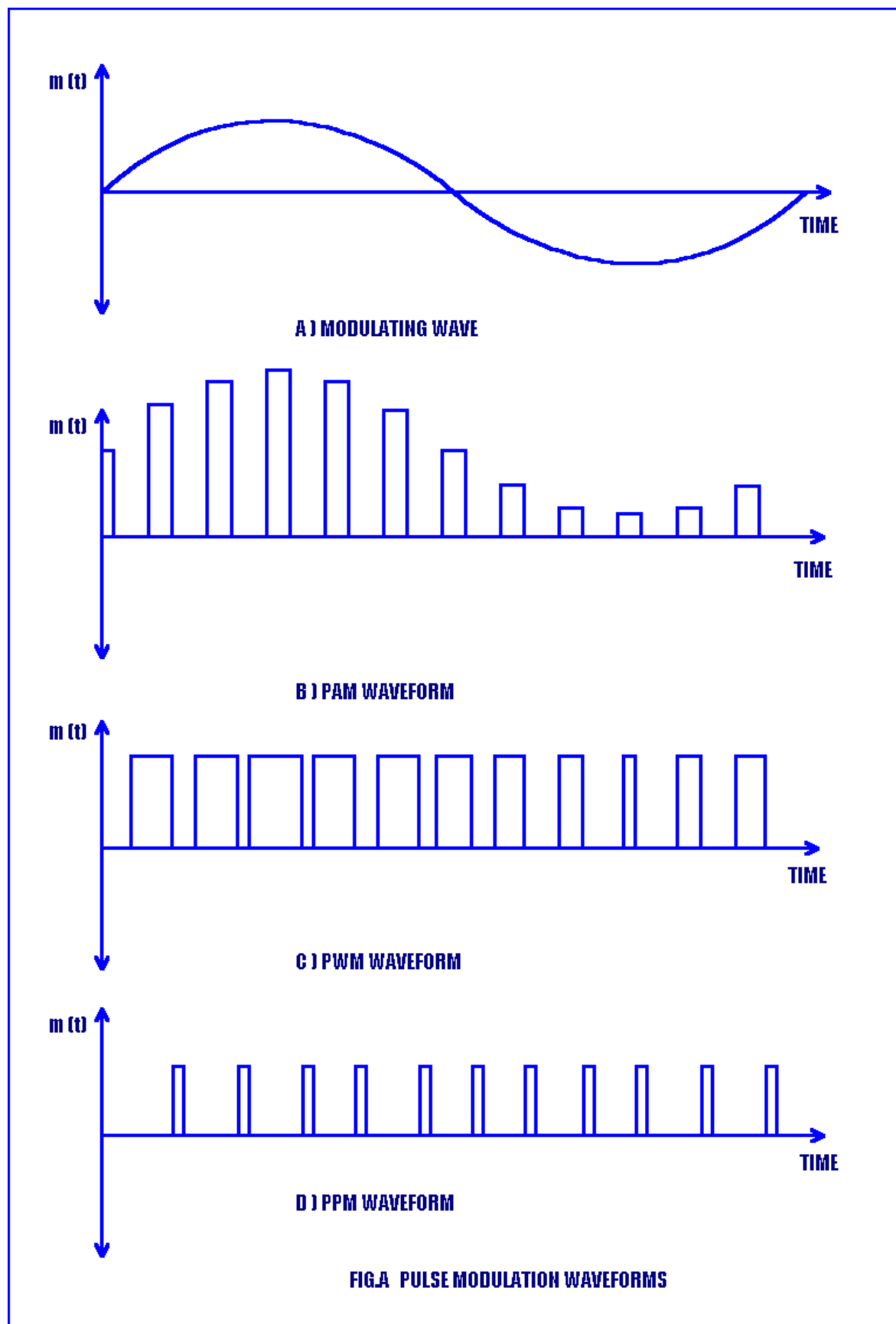
Filter Section

4th order Butterworth Low Pass Filter U9 (KF347) have been provided which filters out the sampling frequency components from the samples and recovers the base band signal.

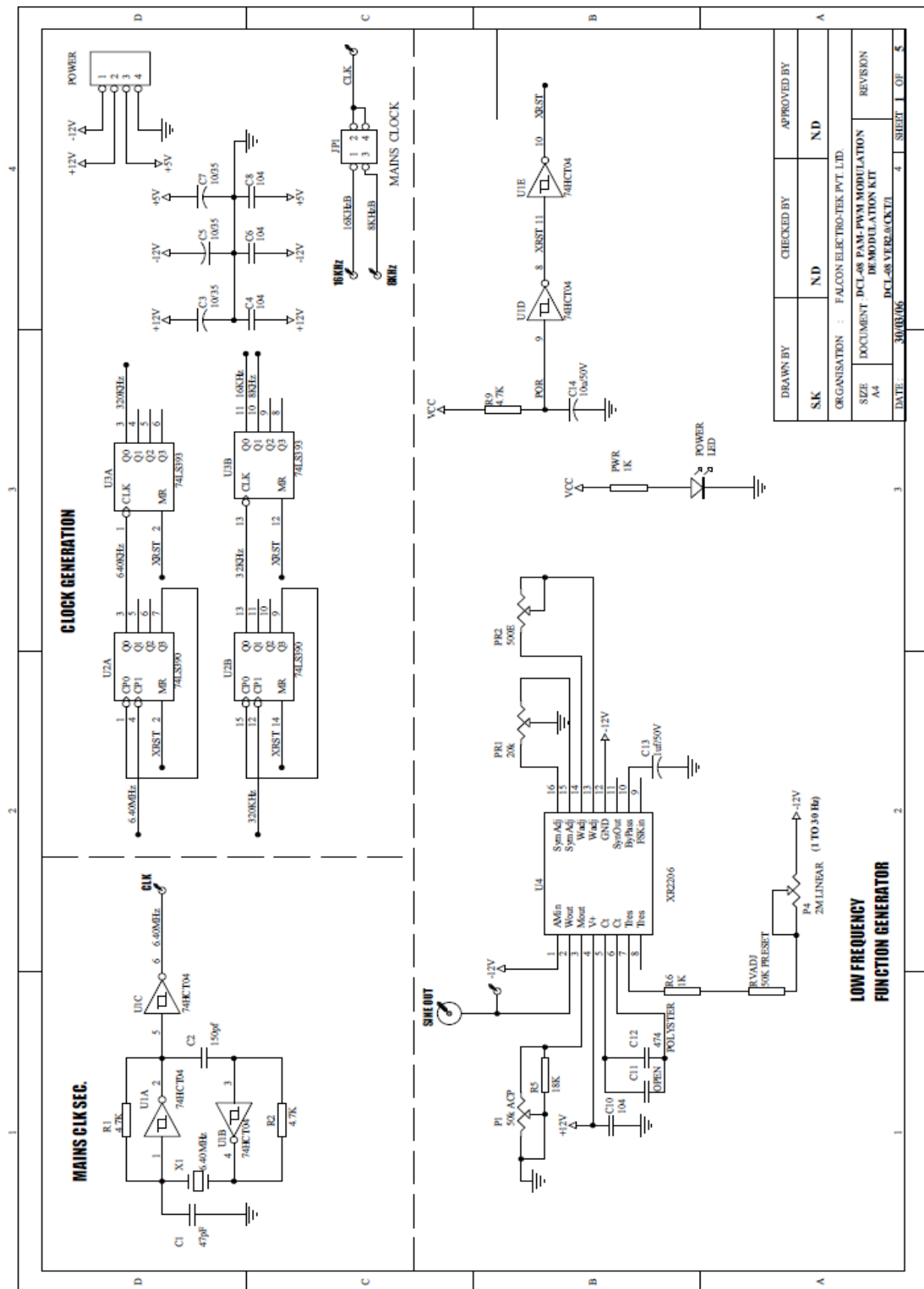
SWITCH FAULTS

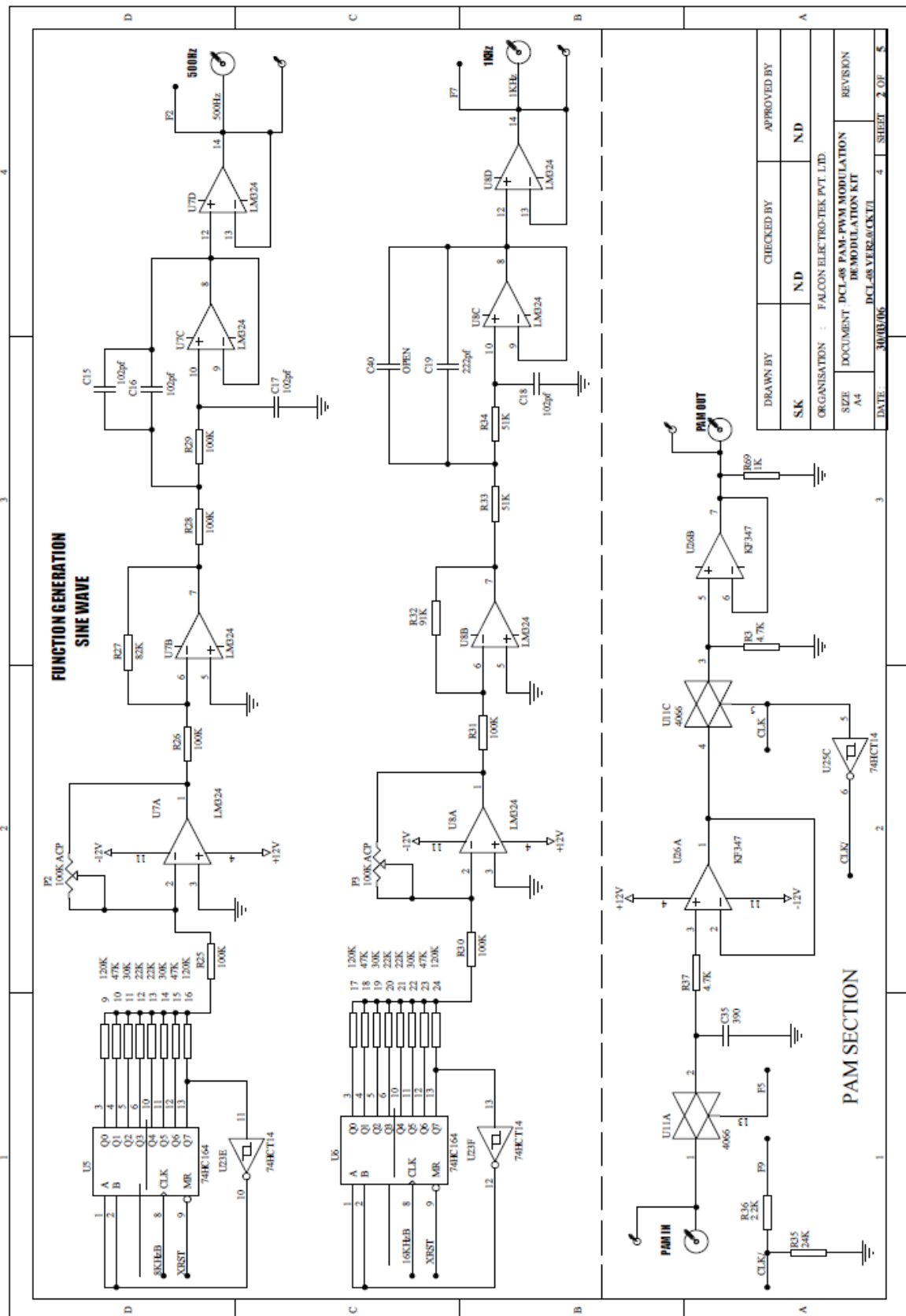
Switch fault section provides 8 switches. These switches can be used to simulate fault conditions in various parts of the circuit. The faults are normally used one at a time. To ensure that the DCL-08 kit is fully operational, all switches should be set to OFF position before use.

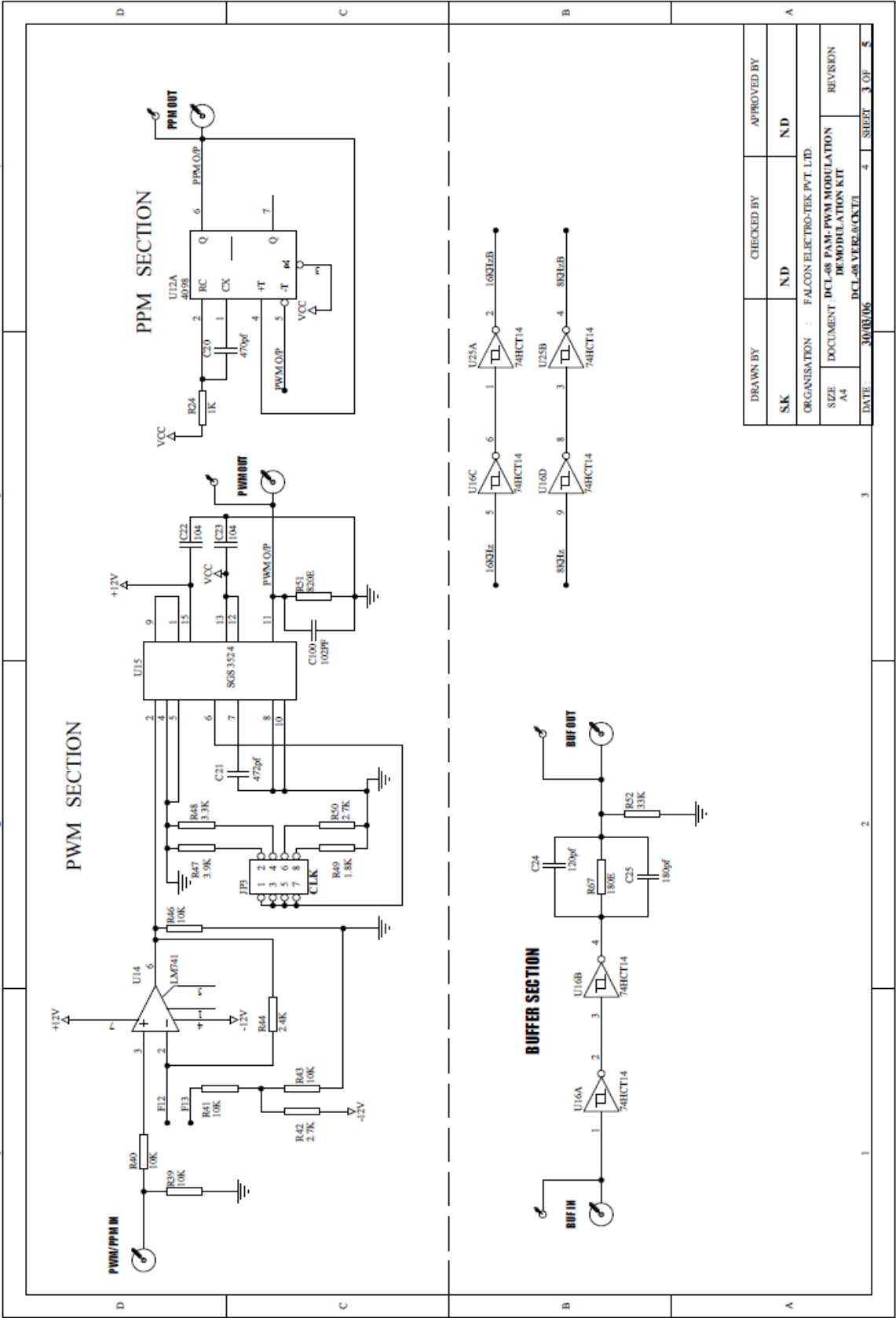
- Put switch **1** of **SF1** in Switch Fault section to **ON** position. The feedback resistor is bypassed from Amplifier section. Gain of Amplifier now depends on potentiometer P5 only.
- Put switch **2** of **SF1** in Switch Fault section to **ON** position. This will generate two mixed sine waves, which could be used as a modulating input signal for modulators PAM, PWM and PPM.
- Put switch **3** of **SF1** in Switch Fault section to **ON** position. This will bypass one filter from filter section. The output consists of ripple with reference to previous output without switch fault.
- Put switch **4** of **SF1** in Switch Fault section to **ON** position. This provides constant high sampling signal to the sampling switch, which in turn gives natural sampling at the output.
- Put switch **5** of **SF2** in Switch Fault section to **ON** position. This removes the control signal of first switch of PAM section, this will open pin of CMOS IC. Due to this output will be abrupt or may follow the input.
- Put switch **6** of **SF2** in Switch Fault section to **ON** position. This opens the connection between PPM Demodulation & Latch enable signal pin of Latch at PPM/PWM demodulator. Demodulated PPM/PWM output is not obtained.
- Put switch **7** of **SF2** in Switch Fault section to **ON** position. This will open MSB from the 8-bit input to DAC at Demodulator section of PWM. Output level reduces in amplitude.
- Put switch **8** of **SF2** in Switch Fault section to **ON** position. This will change Minimum Pulse Width to zero level. Due to this PWM output will be absent for no signal at the PWM input. Minimum pulse width at the output of PWM modulator is not maintained Pulse width will vary according to modulating signal.

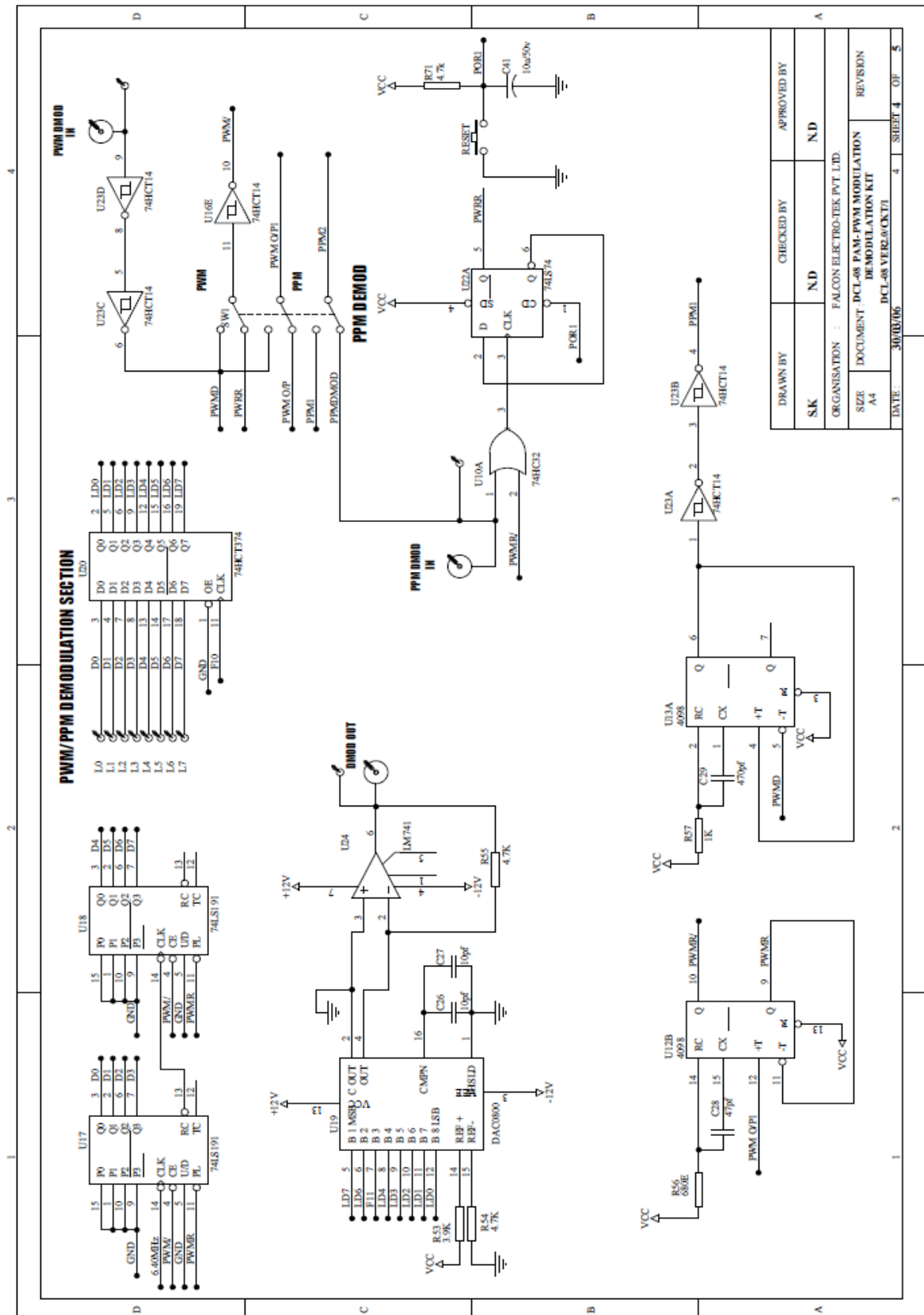


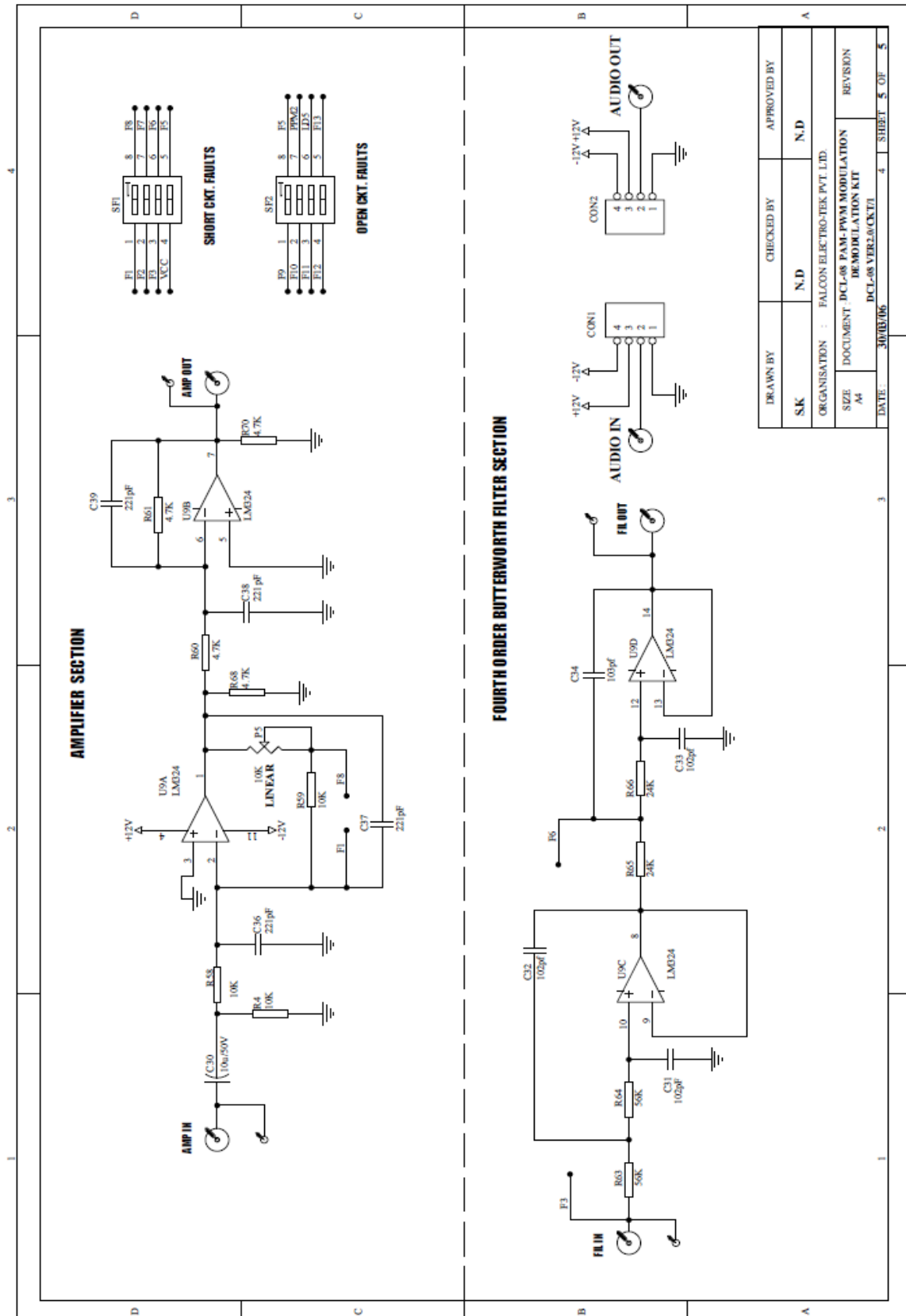
CIRCUIT DIAGRAM



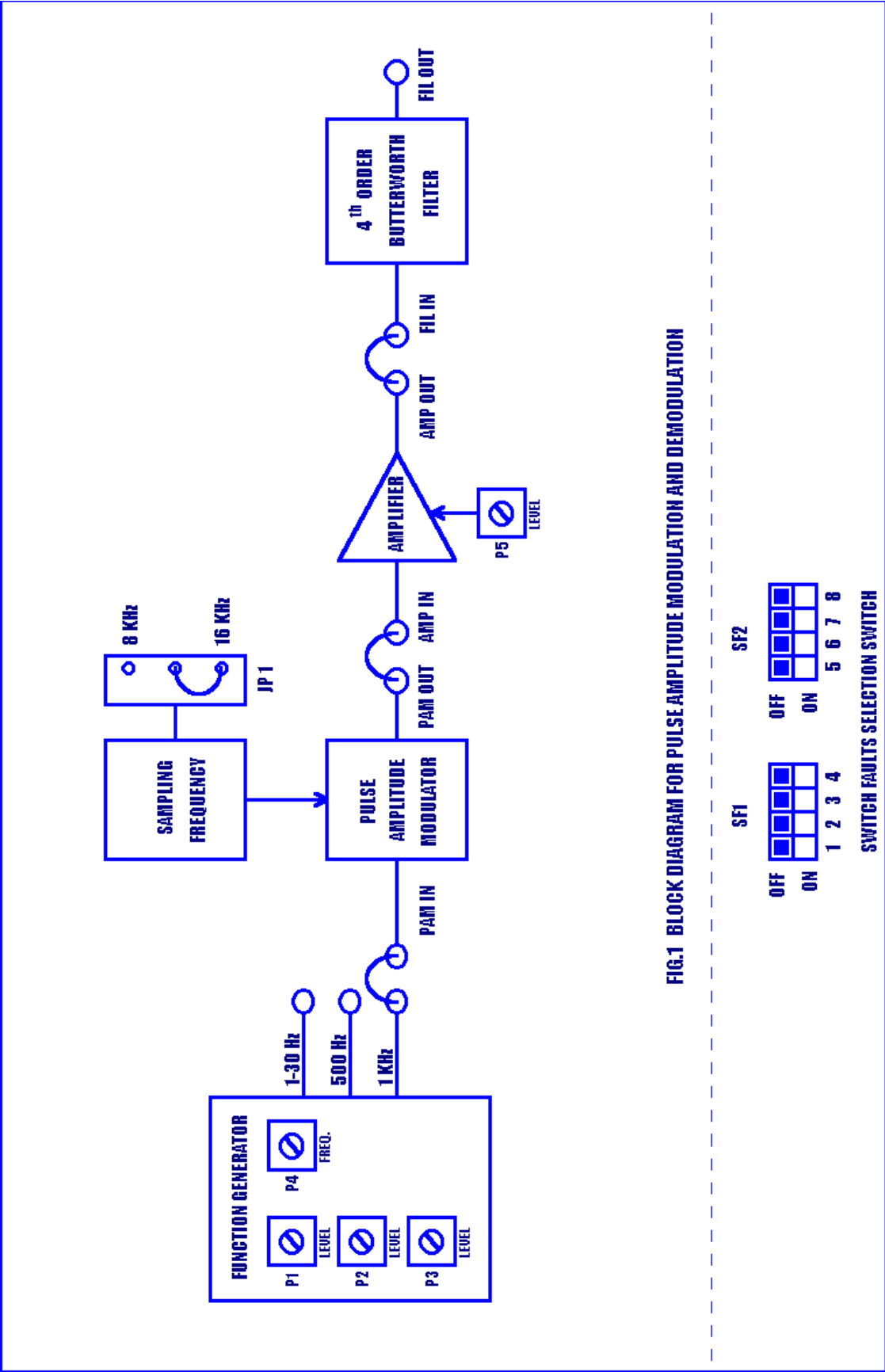








EXPERIMENT NO. 1



EXPERIMENT NO: 1

NAME:

Pulse Amplitude Modulation.

OBJECTIVE:

Study of Pulse Amplitude Modulation & Demodulation

THEORY:

In Pulse Amplitude Modulation, the signal is sampled at regular intervals and the amplitude of each sample is made proportional to the amplitude of the signal at that instant of sampling. This amplitude of each sample is hold for the sample duration to make pulses flat top.

The Pulse Amplitude Demodulator consists of Active Low Pass Butterworth filter. It filters out the sampling frequency and their harmonics from the modulated signal and recovers the base band by integrated action.

EQUIPMENTS:

- DCL-08 Kit.
- Connecting Chords
- Power supply
- Dual trace oscilloscope

NOTE: Keep The Switch Faults In Off Position.

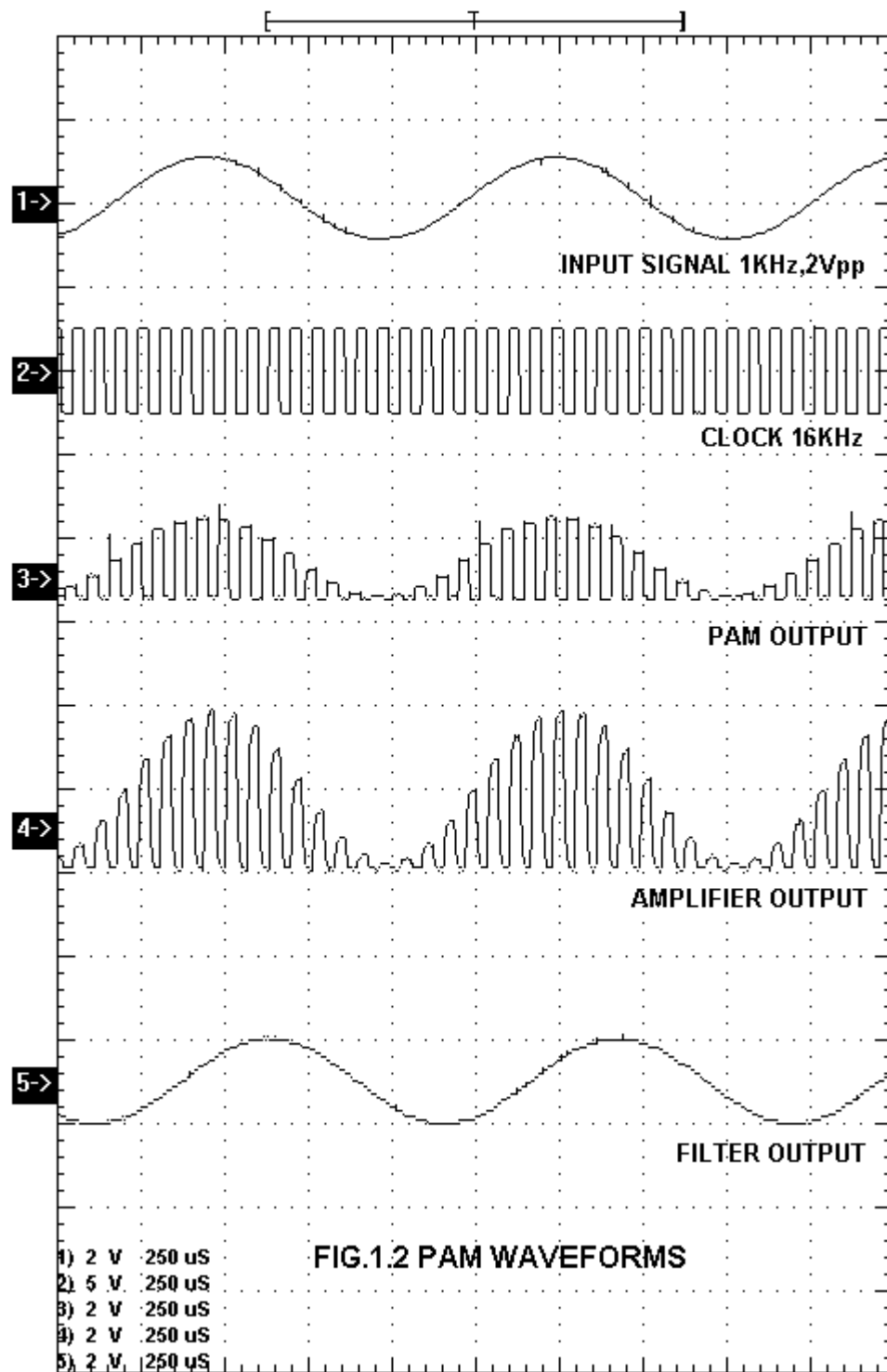
PROCEDURE:

1. Refer to the block diagram (Fig. 1) and carry out the following connections and switch settings.
2. Connect the Power Supply with proper polarity to the kit **DCL-08** and switch it on.
3. Select **16KHz** sampling frequency by jumper **JP1**.
4. Connect the **1KHz**, 2Vp-p sine wave signal generated onboard to **PAM IN** Post.
5. Observe the Pulse Amplitude Modulation output at **PAM OUT** Post.
6. Short the following posts with the Connecting chords provided as shown in block diagram.
PAM OUT and **AMP IN**
AMP OUT and **FIL IN**
7. Keep the amplifier gain control potentiometer **P5** to maximum completely clockwise.
8. Observe the Pulse Amplitude Demodulated signal at **FIL OUT**, which is same as the input signal.
9. Repeat the experiment for different input signal and sampling frequencies.

SWITCHFAULTS:

Note: Keep the connections as per the procedure. Now switch corresponding fault switch button in ON condition & observe the different effect on the output. The faults are normally used one at a time.

- Put switch **1** of **SF1** in Switch Fault section to **ON** position. The feedback resistor is bypassed from Amplifier section. Gain of Amplifier now depends on potentiometer P5 only.
- Put switch **2** of **SF1** in Switch Fault section to **ON** position. This will generate two mixed sine waves, which could be used as a modulating input signal for modulators PAM, PWM and PPM.
- Put switch **3** of **SF1** in Switch Fault section to **ON** position. This will bypass one filter from filter section. The output consists of ripple with reference to previous output without switch fault.
- Put switch **4** of **SF1** in Switch Fault section to **ON** position. This provides constant high sampling signal to the sampling switch, which in turn gives natural sampling at the output.
- Put switch **5** of **SF2** in Switch Fault section to **ON** position. This removes the control signal of first switch of PAM section, this will open pin of CMOS IC. Due to this output will be abrupt or may follow the input.



EXPERIMENT NO. 2

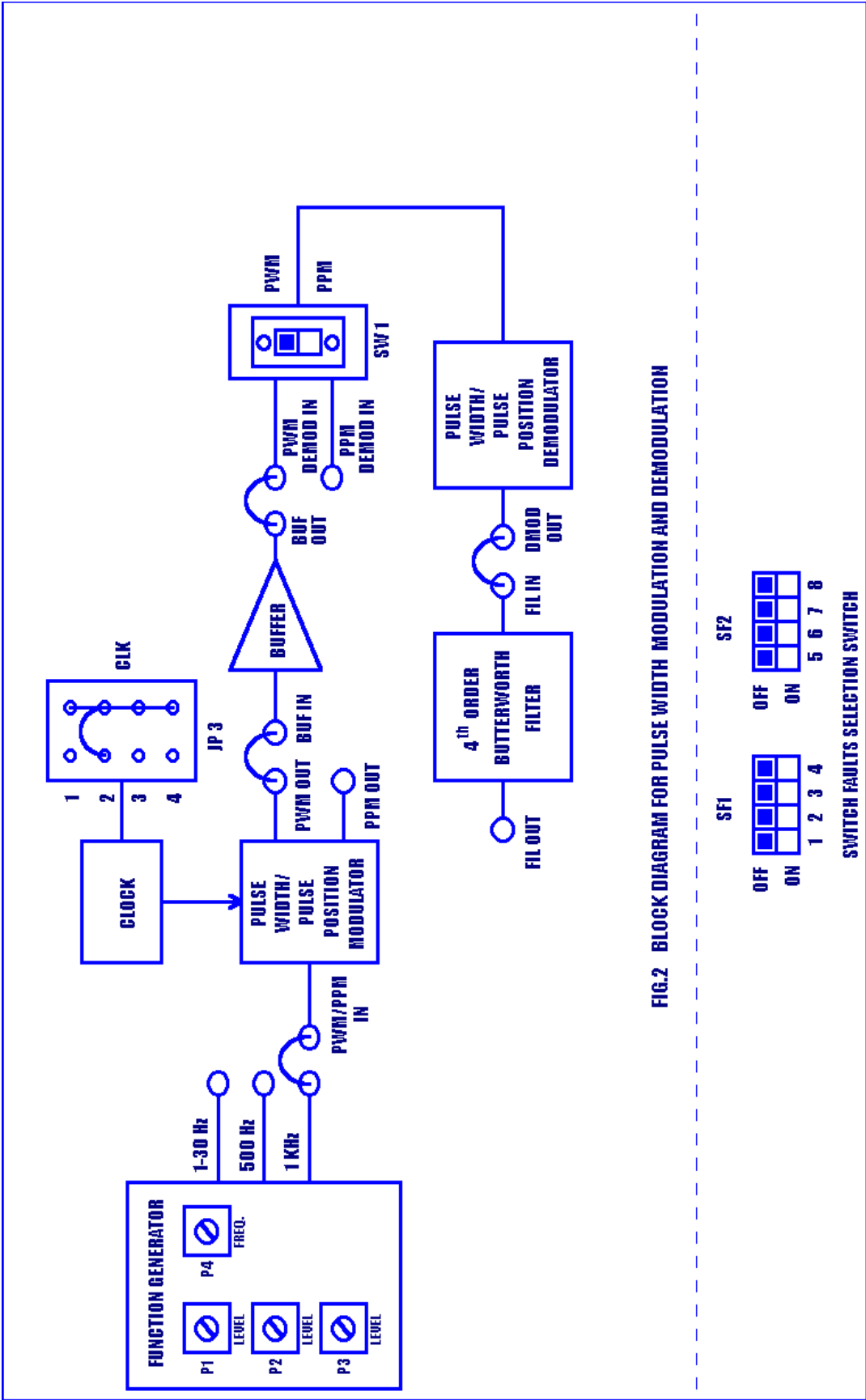


FIG.2 BLOCK DIAGRAM FOR PULSE WIDTH MODULATION AND DEMODULATION

SF1				SF2			
OFF	ON	OFF	ON	OFF	ON	OFF	ON
1	2	3	4	5	6	7	8

SWITCH FAULTS SELECTION SWITCH

EXPERIMENT NO: 2

NAME:

Pulse Width Modulation.

OBJECTIVE:

Study of Pulse Width Modulation and Demodulation

THEORY:

Pulse Width Modulation

This technique of modulation controls the variation of duty cycle of the square wave (With some fundamental frequency) according to the input modulating signal. Here the amplitude variation of the modulation signal is reflected in the ON period variation of square wave. Hence, it is a technique of V to T conversion.

Pulse Width Demodulation

The input signal is Pulse Width Modulated, so the ON time of the signal is changing according to the modulating signal. In this demodulation technique during the ON time of PWM signal one counter is enabled. At the end of ON time, counter gives a particular count, which directly corresponds to the amplitude of input signal. Then this count is fed to a DAC. The output of DAC corresponds to the amplitude of input signal. Thus train of varying pulse widths gives varying count values and accordingly DAC give outputs, which is directly proportional to amplitude of input signal. This is then filtered to get original signal. Thus at the output we get the original modulating signal extracted from PWM wave.

EQUIPMENTS:

- Experimenter kit DCL-08.
- Connecting Chords
- Power supply
- Dual trace oscilloscope

NOTE: Keep The Switch Faults In Off Position.

PROCEDURE:

1. Refer to the block diagram (Fig. 2) and carry out the following connections and switch settings.
2. Connect the Power Supply with proper polarity to the kit **DCL-08** and switch it on.
3. Put jumper **JP3** to 2nd position.
4. Select **1KHZ** 1v-pp sine wave signal generated onboard.
5. Connect this signal to **PWM/PPM IN**.

6. Observe the Pulse Width Modulated output at **PWM OUT** post. Note that since the sampling frequency is high, only blurred band in waveform will be observed due to persistence of vision. In absence of input signal only square wave of fundamental frequency and fixed on time will be observed and no width variation are present. To observe the variation in pulse width, apply 1-30Hz sine wave signal to **PWM/PPM IN** post. Vary the frequency from 1-30 Hz.
7. Short the following posts with the Connecting chords provided as shown in block diagram for demodulation section.
PWM OUT and **BUF IN**
BUF OUT and **PWM DMOD IN**
DMOD OUT and **FIL IN**
8. Observe the Pulse Width Demodulated output at **FIL OUT**.
9. Repeat the experiment for different input signal and different sampling clocks with the help of jumper **JP3**.

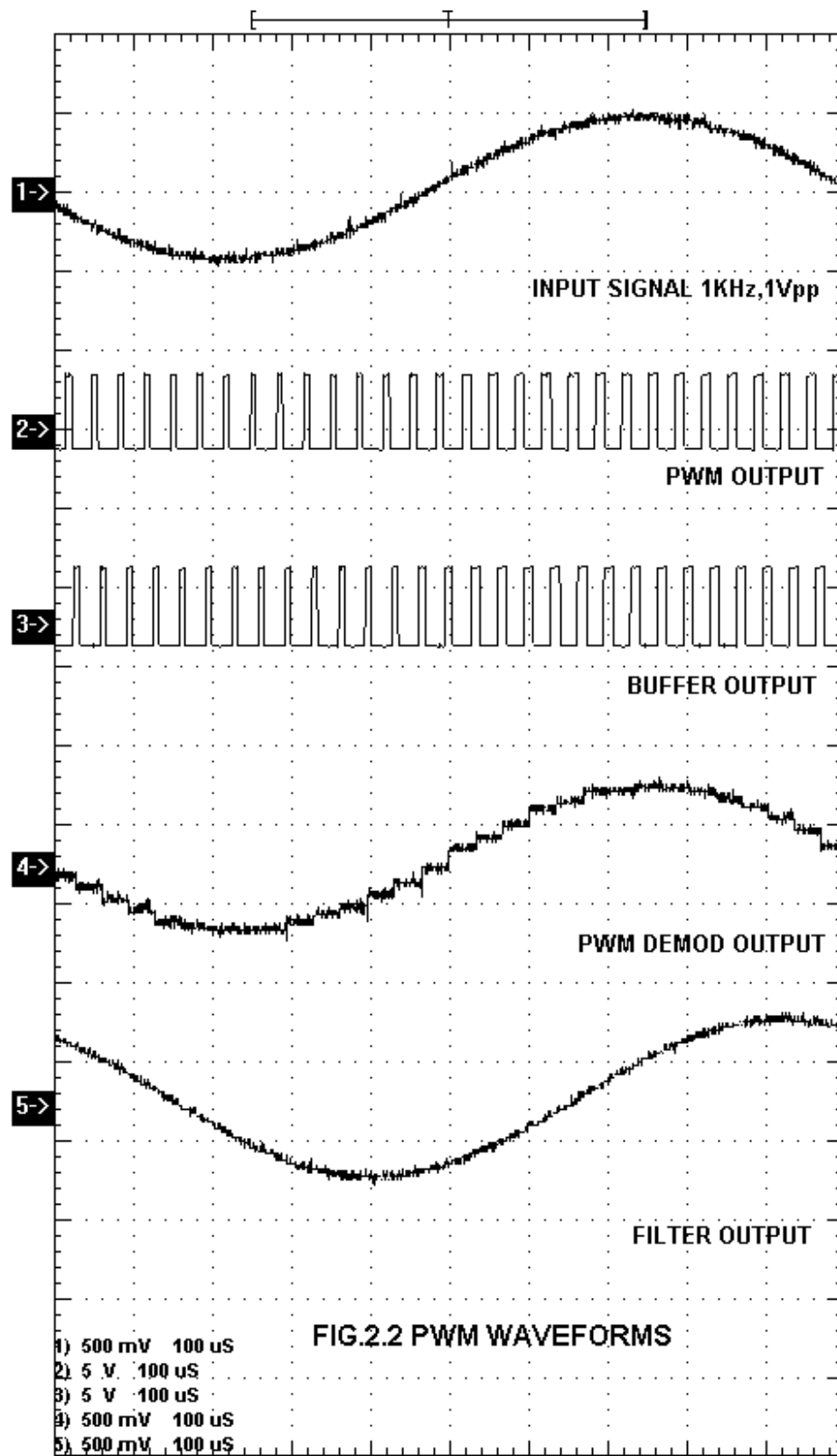
Note: Procedure for Observation of PWM output in DUAL mode:

- Keep CH1 knob of CRO on 1 Volt/ div ac.
- Keep CH2 knob of CRO on 2 Volts/ div ac.
- Keep Times/ div knob on 1 msec.
- Keep the CRO in Dual channel (Auto/ TV mode). Use X10 for expansion.
- After proper triggering of CRO, observe both the signals **PWM IN** and **PWM OUT** simultaneously.

SWITCHFAULTS:

Note: Keep the connections as per the procedure. Now switch corresponding fault switch button in ON condition & observe the different effect on the output. The faults are normally used one at a time.

- Put switch **2** of **SF1** in Switch Fault section to **ON** position. This will generate two mixed sine waves, which could be used as a modulating input signal for modulators PAM, PWM and PPM.
- Put switch **3** of **SF1** in Switch Fault section to **ON** position. This will bypass one filter from filter section. The output consists of ripple with reference to previous output without switch fault.
- Put switch **6** of **SF2** in Switch Fault section to **ON** position. This opens the connection between PPM Demodulation & Latch enable signal pin of Latch at PPM/PWM demodulator. Demodulated PPM/PWM output is not obtained.
- Put switch **7** of **SF2** in Switch Fault section to **ON** position. This will open MSB from the 8-bit input to DAC at Demodulator section of PWM. Output level reduces in amplitude.
- Put switch **8** of **SF2** in Switch Fault section to **ON** position. This will change Minimum Pulse Width to zero level. Due to this PWM output will be absent for no signal at the PWM input. Minimum pulse width at the output of PWM modulator is not maintained Pulse width will vary according to modulating signal.



EXPERIMENT NO. 3

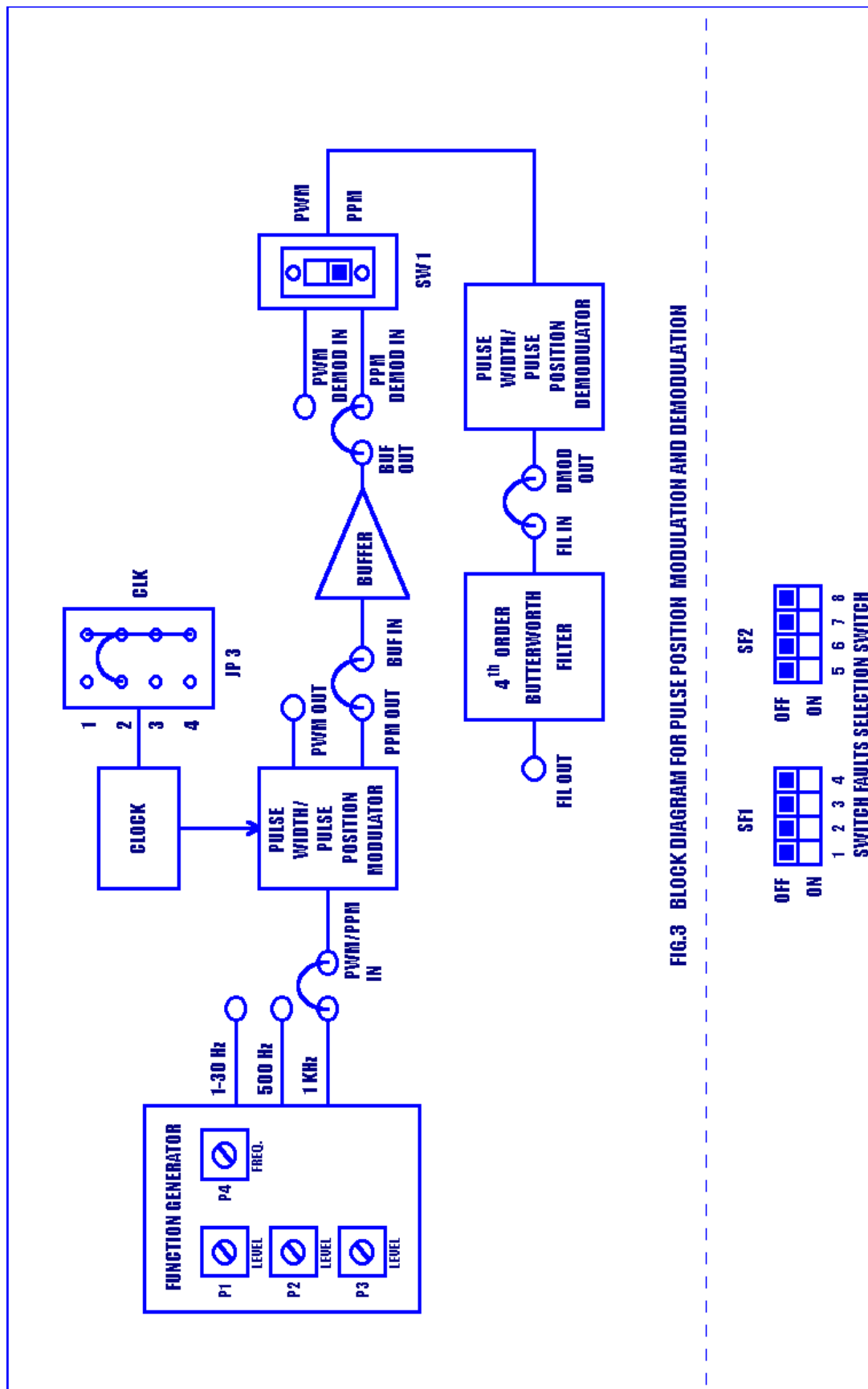


FIG.3 BLOCK DIAGRAM FOR PULSE POSITION MODULATION AND DEMODULATION

EXPERIMENT NO: 3

NAME:

Pulse Position Modulation.

OBJECTIVE:

Study of Pulse Position Modulation and Demodulation

THEORY:

The position of the TTL pulse is changed on time scale according to the variation of input modulating signal amplitude, Width of the pulses and Amplitude of the pulses remain same.

Demodulation

This pulse position modulated signal is converted into PWM pulse form using Monostablemultivibrator. This signal is then demodulated using the same technique of PWM demodulation. In this demodulation technique during the ON time of PWM signal one counter is enabled. At the end of ON time, counter gives a particular count, which directly corresponds to the amplitude of input signal. Then this count is fed to a DAC. The output of DAC corresponds to the amplitude of input signal. Thus train of varying pulse widths gives varying count values and accordingly DAC gives outputs, which is directly proportional to amplitude of input signal. This is then filtered to get original signal. Thus at the output we get the original modulating signal extracted from PWM wave.

EQUIPMENTS:

- DCL-08 Kit.
- Connecting Chords
- Power supply
- Dual trace oscilloscope

NOTE: Keep The Switch Faults In Off Position.

PROCEDURE:

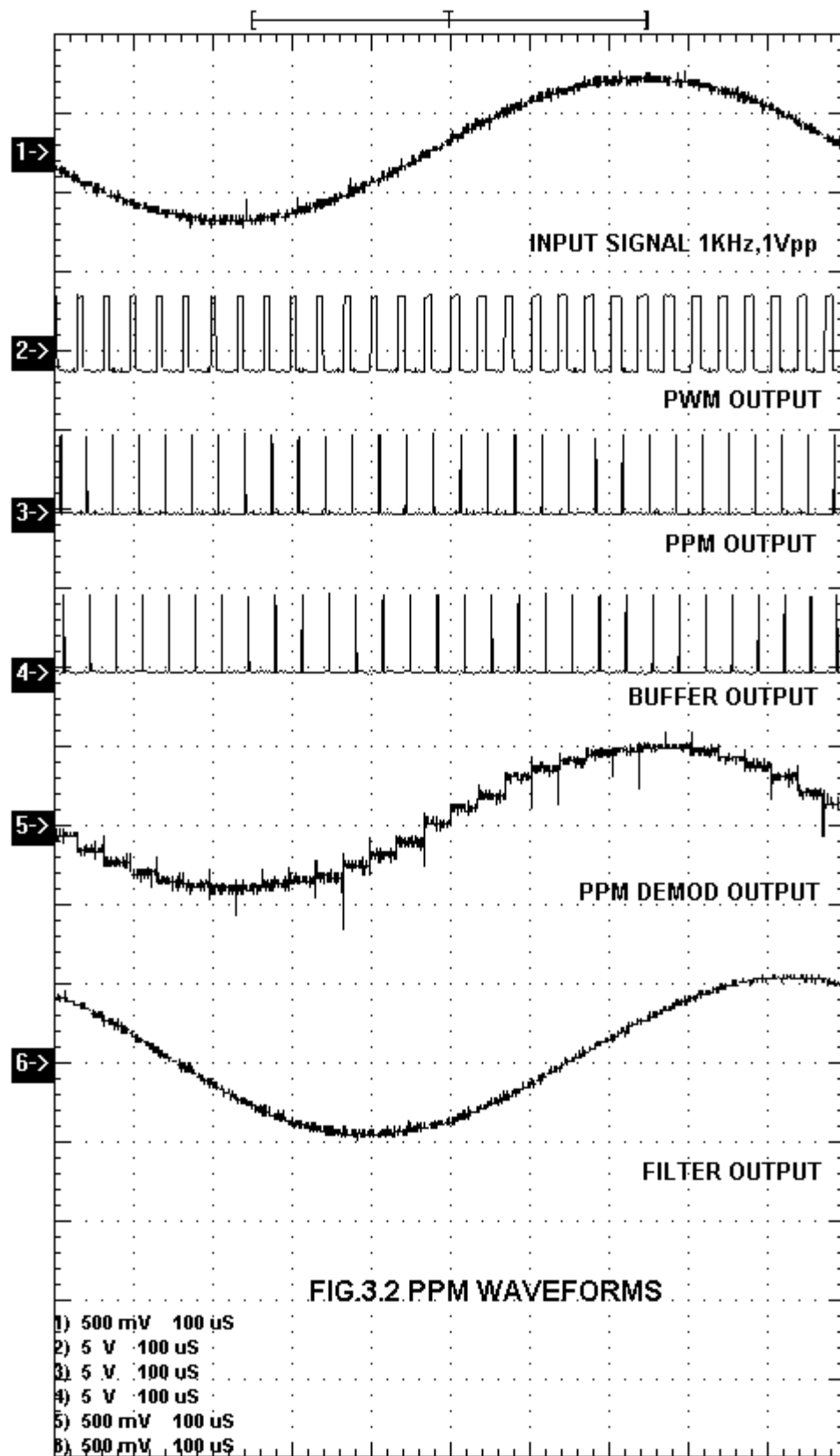
1. Refer to the block diagram (Fig. 3) and carry out the following connections and switch settings.
2. Connect the Power Supply with proper polarity to the kit **DCL-08** and switch it on.
3. Put jumper **JP3** to 2nd position.
4. Select **1KHZ**, 1v-pp sine wave signal generated onboard.
5. Connect the selected signal to the **PWM/PPM IN**.

6. Observe the Pulse Position Modulated output at **PPM OUT** post with shifted position on time scale. Please note amplitude and width of pulse are same and there is shift in position which is proportional to input Analog signal.
7. To observe the variation in pulse positions, apply 1-30Hz sine wave signal to **PWM/PPM IN** post vary the frequency from 1-30 Hz and observe the signal on oscilloscope in dual for posts **PPM OUT** and **PWM OUT** simultaneously.
8. Then short the following posts with the link provided as shown in block diagram for Demodulation section.
PPM OUT and **BUFIN**
BUFOUT and **PPM DMOD IN**
DMOD OUT and **FIL IN**
9. Observe the Pulse Position Demodulated signal at **FIL OUT**.
10. Repeat the experiment at different input signal and different sampling frequencies.

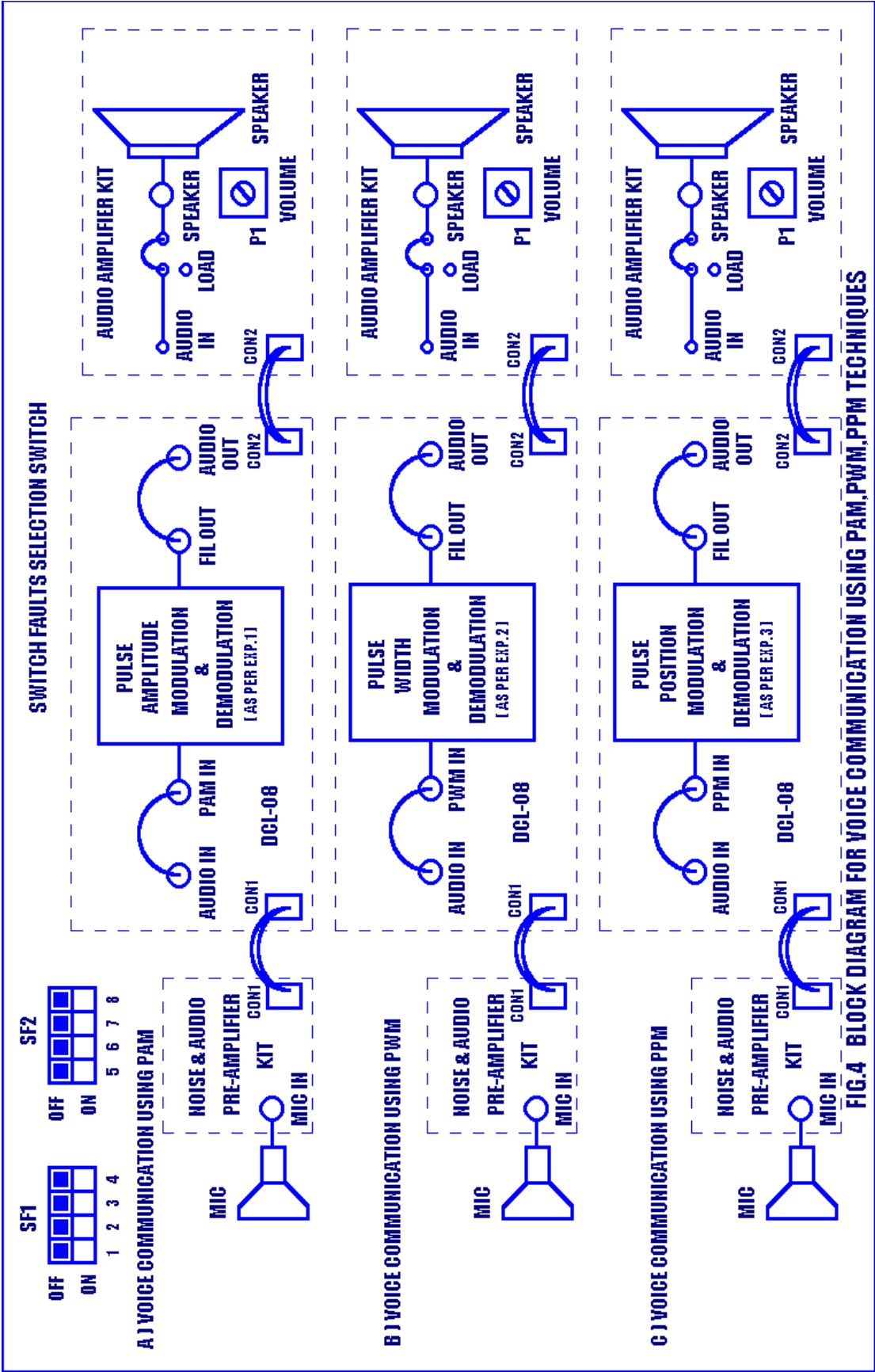
SWITCHFAULTS:

Note: Keep the connections as per the procedure. Now switch corresponding fault switch button in ON condition & observe the different effect on the output. The faults are normally used one at a time.

- Put switch **2** of **SF1** in Switch Fault section to **ON** position. This will generate two mixed sine waves, which could be used as a modulating input signal for modulators PAM, PWM and PPM.
- Put switch **3** of **SF1** in Switch Fault section to **ON** position. This will bypass one filter from filter section. The output consists of ripple with reference to previous output without switch fault.
- Put switch **6** of **SF2** in Switch Fault section to **ON** position. This opens the connection between PPM Demodulation & Latch enable signal pin of Latch at PPM/PWM demodulator. Demodulated PPM/PWM output is not obtained.
- Put switch **7** of **SF2** in Switch Fault section to **ON** position. This will open MSB from the 8-bit input to DAC at Demodulator section of PWM. Output level reduces in amplitude.
- Put switch **8** of **SF2** in Switch Fault section to **ON** position. This will change Minimum Pulse Width to zero level. Due to this PWM output will be absent for no signal at the PWM input. Minimum pulse width at the output of PWM modulator is not maintained Pulse width will vary according to modulating signal.



EXPERIMENT NO. 4



EXPERIMENT NO: 4

NAME:

Voice Communication Using Pulse Modulation Techniques

OBJECTIVE:

Study of Voice communication using PAM, PWM & PPM Techniques

THEORY:

Voice can be modulated, demodulated and reconstructed using any of the modulation techniques such as PAM, PWM & PPM. Before Modulation and Demodulation of Voice signal, set up the basic PAM, PWM or PPM experiment as mentioned earlier and then feed voice signal as input and the demodulated output to speaker.

EQUIPMENTS:

- DCL-08 Kit.
- Connecting Chords
- Power supply
- Dual Trace Oscilloscope
- Microphone
- Noise and Audio Pre-Amplifier kit
- Audio Amplifier kit
- Power connection cables

NOTE: Keep The Switch Faults In Off Position.

PROCEDURE:

1. Refer to the block diagram (Fig. 4) and carry out the following connections and switch settings.
2. Connect the Power Supply with proper polarity to the kit **DCL-08** and switch it on.
3. Connect the NOISE AND AUDIO PRE-AMPLIFIER kit to the **CON1** Post on **DCL-08** kit using power connection cable.
4. Connect the AUDIO AMPLIFIER kit to the **CON2** Post on **DCL-08** kit using power connection cable.
5. Connect Microphone to **MIC IN** post on NOISE AND AUDIO PRE-AMPLIFIER kit.
6. Connect the input signal from **AUDIO IN** Post to **PAM IN** Post.
7. Short the following posts with the link provided as shown in block diagram.
PAM OUT and **AMP IN**
AMP OUT and **FIL IN**
8. Connect the output signal from **FIL OUT** to **AUDIO OUT**.

9. Observe Voice communication through PAM at various test posts.
10. Similarly repeat voice link through **PWM&PPM** techniques by referring the procedure of experiments 2 & 3.

SWITCHFAULTS:

Note: Keep the connections as per the procedure. Now switch corresponding fault switch button in ON condition & observe the different effect on the output. The faults are normally used one at a time.

- Put switch **1** of **SF1** in Switch Fault section to **ON** position. The feedback resistor is bypassed from Amplifier section. Gain of Amplifier now depends on potentiometer P5 only.
- Put switch **3** of **SF1** in Switch Fault section to **ON** position. This will bypass one filter from filter section. The output consists of ripple with reference to previous output without switch fault.
- Put switch **4** of **SF1** in Switch Fault section to **ON** position. This provides constant high sampling signal to the sampling switch, which in turn gives natural sampling at the output.
- Put switch **5** of **SF2** in Switch Fault section to **ON** position. This removes the control signal of first switch of PAM section, this will open pin of CMOS IC. Due to this output will be abrupt or may follow the input.
- Put switch **6** of **SF2** in Switch Fault section to **ON** position. This opens the connection between PPM Demodulation & Latch enable signal pin of Latch at PPM/PWM demodulator. Demodulated PPM/PWM output is not obtained.
- Put switch **7** of **SF2** in Switch Fault section to **ON** position. This will open MSB from the 8-bit input to DAC at Demodulator section of PWM. Output level reduces in amplitude.
- Put switch **8** of **SF2** in Switch Fault section to **ON** position. This will change Minimum Pulse Width to zero level. Due to this PWM output will be absent for no signal at the PWM input. Minimum pulse width at the output of PWM modulator is not maintained Pulse width will vary according to modulating signal.